Descriptive written Test $Co_1 - AT_3$

1. Smart Home Automation system using intelligent Agents

Introduction :-

An intelligent agent is a system that observes its environment through sensors and takes suitable actions using actuators to achieve specific objectives, in a smart home automation system, sensors detect motion, temperature, light, and time, while actuators control lights, air conditioners, and security devices.

The four masar types of intelligent agents are simple reflex.

1. Simple Reflex Agent :-

A simple reflex agent selects actions based only on the current input using condition action rules.

Working in a smart home :-

=> If motion is detected, turn on the light.

=> If the temperature exceeds 28°C, turn on the air conditioner.

=> If an unauthorized person is detected, activate the alarm.

Advantages :- simple, Fast, and easy to implement.

Disadvantages :- It does not remember past ever or learn user perferences. It may make onsui decisions when the environment changes.

2. Model-Based Agent

A model-based agent maintains an internal state of the environment and uses previous observations.

Working in a smart home :-

=> Remembers whether a room is occupied.

=> Tracks the current temperature and previous temperature changes.

=> Checks whether the home is in sipping mode or away mode before controlling devices.

Advantages :- Balances Handles Partially observable environments and makes better decisions using stored information.

Disadvantages :- Requires more memory and a model of the environment.

3. Goal - Based Agent :

A goal-based agent selects actions that help achieve a specific goal.

working in a smart home :-

=> Achieves the goal of maintaining a comfortable room temperature.

=> Ensures that doors are locked when the home is unoccupied.

Advantages :- can Plan actions and choose different Paths to achieve a goal.

Disadvantages :- Planning may require additional computation and time.

4. **Utility - Based Agent :-** A utility - based agent evaluates Possible actions using a utility function and selects the action that Provides the overall benefit.

Working in a Smart home :-

=> Balance comfort with electricity consumption.

=> Adjusts air conditioning based on temperature,

=> Prioritizes security while avoiding unnecessary alarms.

Advantages :- Balance multiple objectives and makes flexible decisions.

Disadvantages :- Designing an appropriate utility function can be complex.

2. **Robot Navigating an unknown maze using UCS and IDS**

Introduction :- A Robot navigating an unknown maze must discover the available Paths, avoide obstacles, and reach the texit without knowing the complex layout in advance uniformed search algorithms explore Possible states without using additional goal.

Two important uniformed search strategies are uniform cost search (UCS) and Iterative Deepening search (IDS)

1. Uniform Cost Search (UCS) :- Uniform cost search expands the node with the lowest total path cost from the starting position. It uses a priority queue to select next node.

Working :-

* Place the starting position in the priority queue cost zero.
* Remove the node with the lower cost.
* Check whether it is the exit.
* If it is not the exit, explore its unvisited neighboring positions.
* Repeat until the exit is found or no reachable remain.

Example :- If one route to the exit costs 12 units and another costs 7 units, UCS finds the lower-cost route,

Advantages :-

=> Finds the least-cost path when step costs are non-negative.

=> Works well when different paths have different costs

=> is complete under standard stats conditions, such as a finite branching factor and a positive minimum step cost.

Disadvantages :-

=> may explore many unnecessary positions.

=> Required substantial memory to store the frontier and explored nodes.

=> Can be slow in large mazes.

Time and Space complexity :- For a general search tree, UCS can require exponential time and space. Its complexity depends on the branching factor.

2. **Iterative Deepending Search (IDS) :-**

Iterative deepening search combines the space efficiency of depth-first search with the completeness of breadth-first for unit-cost paths.

Working :-

1. Start with a depth limit of zero.
2. Search for the exit within that limit
3. If the exit is not found, increases the limit to one.
4. Repeat the search with progressively larger limit.
5. Stop when the exit is found.

Example :- The robot first checks positions at depth 0, then depth 1, depth 2, and so on until it reaches.

Advantages :-

=> Uses much less memory than BFS.

=> Is complete when the branching factor is finite.

=> Finds a shallowest solution when all action have equal cost.

Disadvantages :-

=> Repeats searches of previously visited.

=> Can take a long time in mazes.

=> Does not guarated the least-path cost.

Time and Space Complexity :-

* Time $O(b^d)$
* Space $O(bd)$

Here, b is the branching factor, d is the depth of the shallowest solution.